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(57) Abstract :
The present invention provides an efficient prime bit Vedic multiplier for double precision floating point numbers. The Prime bit Vedic multiplier is a multiplication algorithm that has gained attention in Digital Signal Processing (DSP) due to its ability to provide efficient and fast multiplication of floating-point numbers. The algorithm is based on Vedic mathematics and is designed to consume less area and provide less delay compared to other multiplication algorithms. The Prime bit Vedic multiplier is particularly useful in double-precision floating-point multiplication to multiply the mantissa. This step is essential in DSP algorithms, and the prime bit Vedic multiplier can significantly improve the overall performance and efficiency. The algorithm is verified in Verilog HDL and can be simulated and synthesized using Xilinx Vivado 2022.2 software, which can optimize the algorithm for hardware implementation.

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